ACC250: Intro to Financial Accounting

# Ch6. Merchandising Operations & Multistep Income Statements

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# Learning Objectives

- Apply the **inventory equation** linking **inventory** and **cost of goods sold**.
- Compare the **periodic** and **perpetual** inventory systems.
- Record purchases and sales, including **discounts**, **returns**, and **transportation costs**.
- Prepare a **multistep income statement** and compute the **gross profit ratio**.

# Table of Contents


1. Introduction to Inventory

2. Inventory System and Journal Entries
   - Periodic Inventory System
   - Perpetual Inventory System
   - Inventory Purchase Transactions
   - Inventory Selling Transactions

3. Multistep Income Statements and Profit Ratios

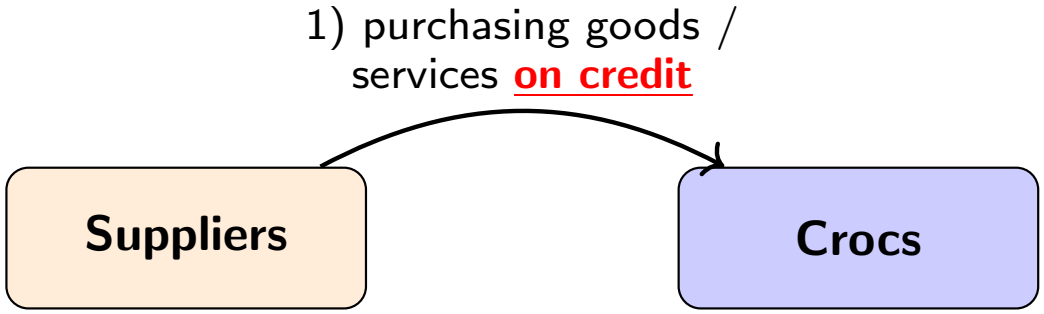
# Operating Cycle




**Operating cycle**:

- the period from the purchase of goods and services to the collection of cash from customers.[1]

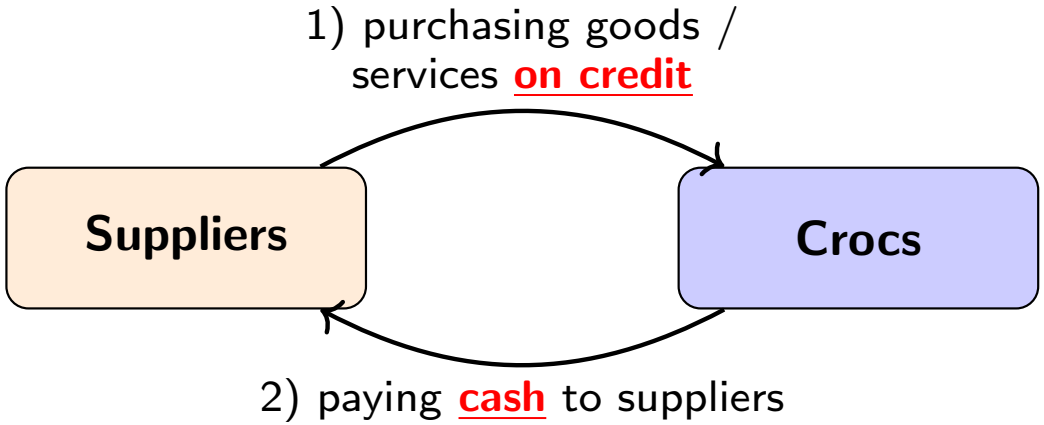
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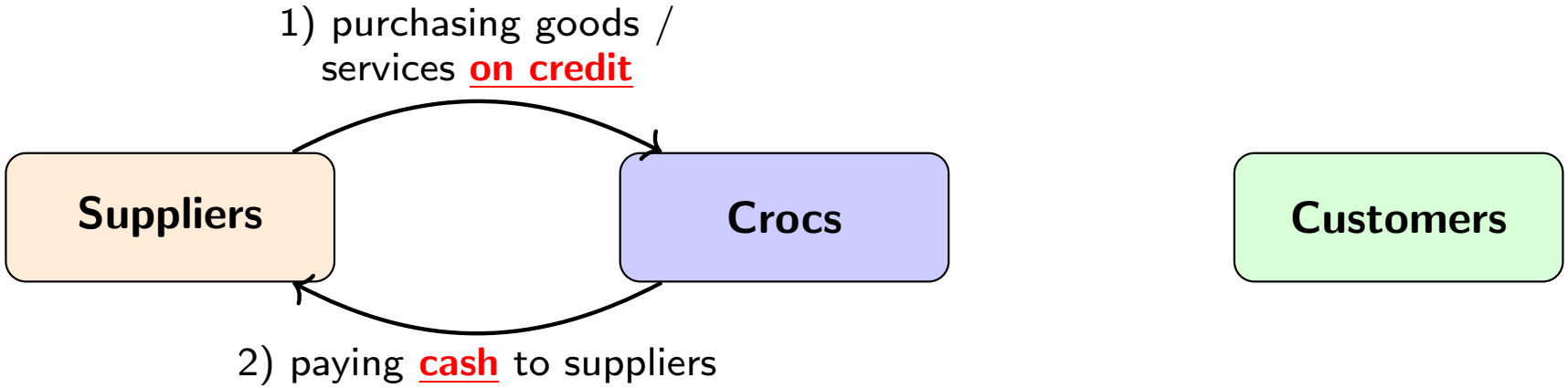
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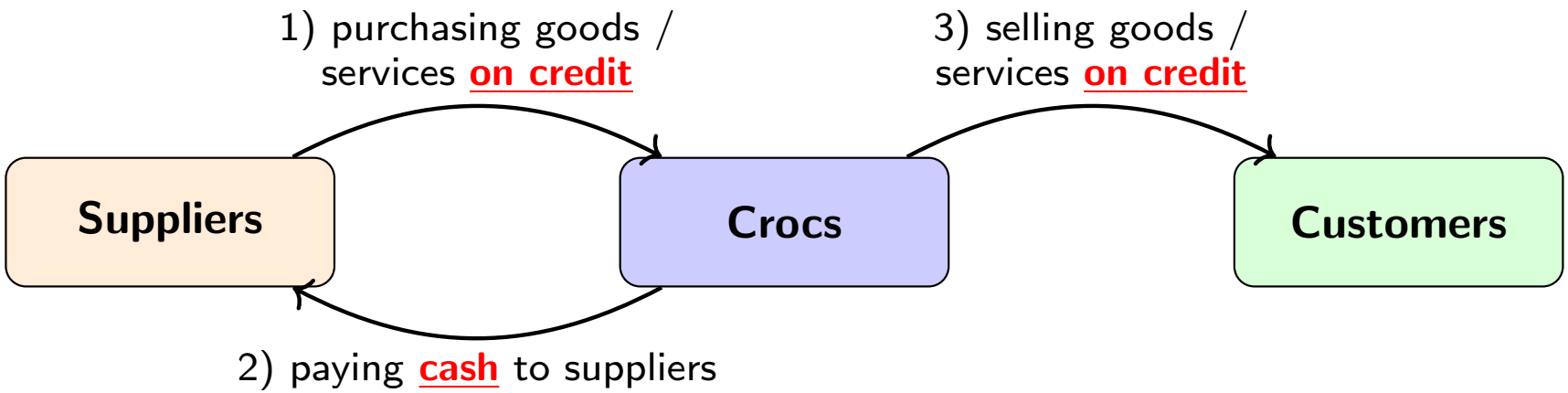
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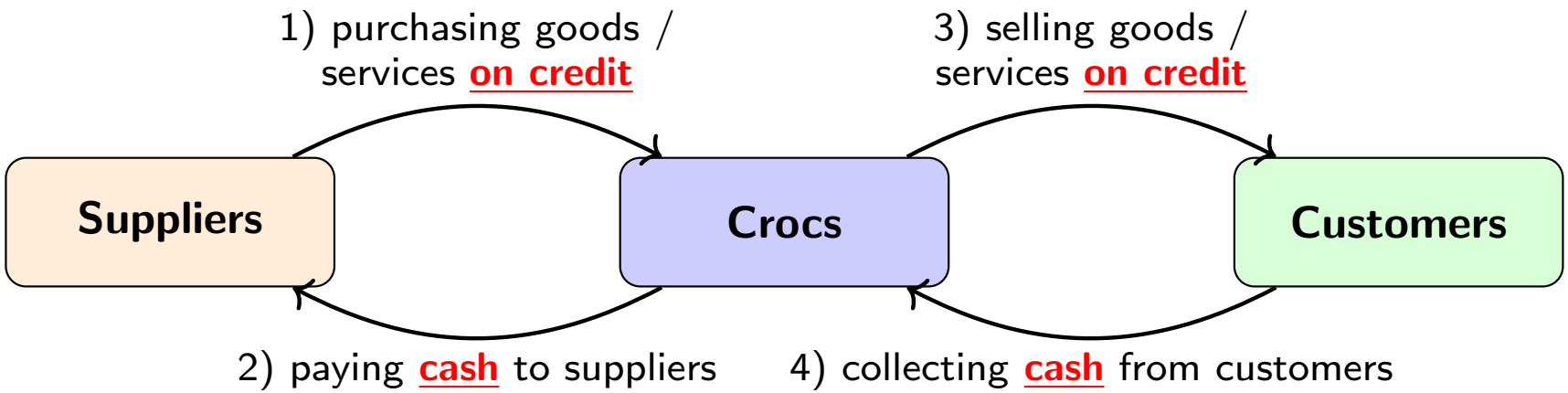
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[1] ACC250 covers **merchandising** firms only (e.g., Amazon, Walmart, Target): they buy finished goods and resell them. Manufacturing firms (e.g., Apple, Tesla, GM) are covered in ACC255.

# Inventory - Concept

## Inventory (Asset)

The total value of the goods held for sale in the ordinary course of business.

## Cost Principle

Assets are initially recorded at their **acquisition cost**, which includes all costs necessary to acquire the asset and prepare it for its intended use.

# Inventory - Concept

## Inventory (Asset)

The total value of the goods held for sale in the ordinary course of business.

- Inventory is a **current asset**.
- Inventory is recorded at its **cost** (under the **cost principle**).
  (i.e., how much the company paid to acquire the inventory.)

## Cost Principle

Assets are initially recorded at their **acquisition cost**, which includes all costs necessary to acquire the asset and prepare it for its intended use.

# Inventory Calculation Example

**EX.** How much inventory was sold (i.e., **Cost of Goods Sold, COGS**) given the following information?

- At the end of the last year, the inventory balance was $100.
- In January, your company purchased $600 additional inventories (i.e., Purchases).
- At the end of January, you found that the inventory balance (i.e., after selling some of it) is $300.

**Answer:**

- **Goods available for sale**:
- **Cost of Goods Sold**:

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**Answer:**

- **Goods available for sale**: $\$100 + \$600 = \$700$
- **Cost of Goods Sold**:

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**Answer:**

- **Goods available for sale**: $100 + $600 = $700
- **Cost of Goods Sold**: $700 - $300 = $400

# Inventory - Equation

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End. Inventory = Beg. Inventory + Purchases - Cost of Goods Sold

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Inventory is

- increased when purchased;
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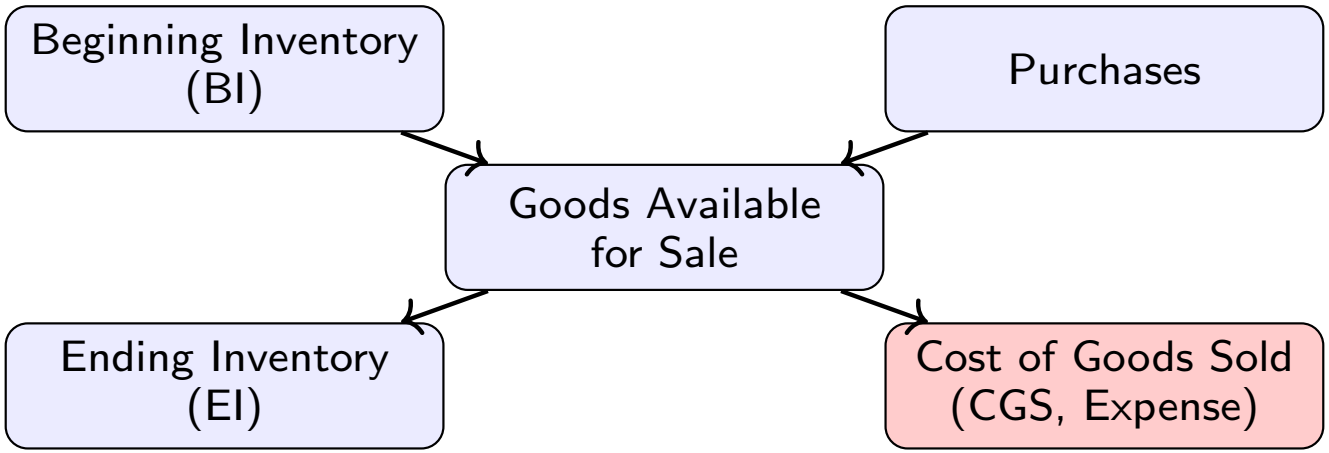
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# Inventory Equation - to get EI

**EX 1.** Suppose a company has the following information for the year:

- Beginning Inventory (BI): \$2K
- Purchases: \$8,500
- Cost of Goods Sold: \$8,200

What is the Ending Inventory (EI) for the year?

- **Goods Available for Sale**:
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- **Goods Available for Sale**: \$2K + \$8,500 = \$10,500
- **Ending Inventory**: \$10,500 - \$8,200 = \$2,300

# Inventory Equation - to get CGS

**EX 2.** Suppose a company has the following information for the year:

- Beginning Inventory (BI): \$2K
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What is the Cost of Goods Sold (CGS) for the year?[2]

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[2] This calculation is required under Periodic Inventory System.

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- Beginning Inventory (BI): \$2K
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What is the Cost of Goods Sold (CGS) for the year?[2]

- **Goods Available for Sale**: \$2K + \$8,500 = \$10,500
- **Cost of Goods Sold**: \$10,500 - \$2,300 = \$8,200

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# Inventory vs Supplies

**Similarities:**

- Both Inventory and Supplies are **assets**.
- Both are **current assets**.
- Both are purchased from **suppliers**.
- Both are usually recorded with **Accounts Payable**.

**Difference:**

- **Inventory** is the goods that a company has purchased but not yet _____.
- **Supplies** are the goods that a company has purchased but not yet _____.

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**EX.** The company purchased $50 inventory on account:

| | | |
|---|---|---|
| Inventory (+A) | 50 | |
| Accounts Payable (+L) | | 50 |

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| Accounts Receivable (+A) | 100 | |
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| Inventory (-A) | | 50 |

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**The two sets of entries are recorded when the inventory is sold!**

# Relationship between Assets and Expenses

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- Assets are resources that a company owns or controls.
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Assets are turned into expenses when they are used up to generate revenues.

- **Inventory** is turned into **Cost of Goods Sold** when it is sold.
- **Supplies** are turned into **Supplies Expense** when they are used.
- **Equipment** is turned into **Depreciation Expense** when it is used.
- **Prepaid Rent** is turned into **Rent Expense** when it is used.

# Accounts Payable Terms (e.g., 2/30, n/60)

Inventory and Accounts Payable:

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- **2/30**: the buyer can get a _____ discount if they pay within _____ days.
- **60**: the buyer should pay the full amount within _____ days without any discount.

**Note:** The disctinction between "n" vs "g" goes beyond the scope of this course. "n" means "net" method; "g" means "gross" method. **Net method** means that the buyer should pay the full amount without any discount. **Gross method** means that the buyer should pay the full amount with the discount.

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# Example

**EX 1.** A company purchases \$5K of inventory on account from a supplier with terms **2/30, n/60**. If the company pays the invoice within 30 days, how much should be paid?

**EX 2.** What if the company pays the invoice 45 days after the purchase?

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\$4,900 (after 2% discount)

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\$5K (no discount)

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# Inventory System

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1. **Periodic inventory system**: Inventory is physically counted only at the end of reporting period.[3]
2. **Perpetual inventory system**: Inventory is updated perpetually (i.e., every time inventory is bought, sold, or returned).
   $\rightarrow$ More accurate and modern when automized inventory tracking is available.

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# Periodic Inventory System: Example

Under the periodic system, **CGS** is **inferred**: **CGS** = **BI** + **Purchases** − **EI**.

**EX 1.** Our company counts inventory at the end of each month. The following information is available:

- Beginning Inventory (BI): $100
- Purchases: $200
- Ending Inventory (EI): $150

What is the Cost of Goods Sold (CGS) for the month?

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- CGS: **$150**

# Perpetual Inventory System

Inventory is updated perpetually (i.e., every time inventory is bought, sold, or returned).

**EX 1.** BI= \$100. Consider the following transactions. What is the inventory balance at the end of each month? and what is the CGS? (\$ amounts are **acquisition costs.**)

| Date | Purchases | Sales | Inventory Balance |
|---|---|---|---|
| Jan 1 | \$300 | | |
| Jan 10 | \$100 | | |
| Jan 20 | | \$200 | |

- EI:
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| Jan 1 | \$300 | | \$400 |
| Jan 10 | \$100 | | \$500 |
| Jan 20 | | \$200 | **\$300** |

- EI: **\$300**
- CGS: **\$200**

**EX 2.** BI= \$500. Consider the following transactions. What is the inventory balance at the end of each month? and what is the CGS? (\$ amounts are **acquisition costs.**)

| Date | Purchases | Sales | Inventory Balance |
|---|---|---|---|
| Feb 1 | \$200 | | |
| Feb 5 | | \$100 | |
| Feb 10 | \$100 | | |
| Feb 15 | | \$200 | |
| Feb 28 | | \$200 | |

- EI: _____
- CGS: _____

Once again, under Perpeutal Inventory System,

- Inventory and CGS are updated every time inventory is bought and sold.
- Inventory equation is **not needed** to calculate the CGS.

# Perpetual Inventory System

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| Date | Purchases | Sales | Inventory Balance |
|---|---|---|---|
| Feb 1 | \$200 | | \$700 |
| Feb 5 | | \$100 | |
| Feb 10 | \$100 | | |
| Feb 15 | | \$200 | |
| Feb 28 | | \$200 | |

- EI: _____
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# Perpetual Inventory System

**EX 2.** BI= \$500. Consider the following transactions. What is the inventory balance at the end of each month? and what is the CGS? (\$ amounts are **acquisition costs.**)

| Date | Purchases | Sales | Inventory Balance |
|---|---|---|---|
| Feb 1 | \$200 | | \$700 |
| Feb 5 | | \$100 | \$600 |
| Feb 10 | \$100 | | \$700 |
| Feb 15 | | \$200 | |
| Feb 28 | | \$200 | |

- EI: _____
- CGS: _____

Once again, under Perpeutal Inventory System,

- Inventory and CGS are updated every time inventory is bought and sold.
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| Feb 28 | | \$200 | |

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|---|---|---|---|
| Feb 1 | \$200 | | \$700 |
| Feb 5 | | \$100 | \$600 |
| Feb 10 | \$100 | | \$700 |
| Feb 15 | | \$200 | \$500 |
| Feb 28 | | \$200 | \$300 |

- EI: _____
- CGS: _____

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| Feb 5 | | \$100 | \$600 |
| Feb 10 | \$100 | | \$700 |
| Feb 15 | | \$200 | \$500 |
| Feb 28 | | \$200 | \$300 |

- EI: **300**
- CGS: _____

Once again, under Perpeutal Inventory System,

- Inventory and CGS are updated every time inventory is bought and sold.
- Inventory equation is **not needed** to calculate the CGS.

# Perpetual Inventory System

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|---|---|---|---|
| Feb 1 | \$200 | | \$700 |
| Feb 5 | | \$100 | \$600 |
| Feb 10 | \$100 | | \$700 |
| Feb 15 | | \$200 | \$500 |
| Feb 28 | | \$200 | \$300 |

- EI: **300**
- CGS: **500**

Once again, under Perpeutal Inventory System,

- Inventory and CGS are updated every time inventory is bought and sold.
- Inventory equation is **not needed** to calculate the CGS.

# Note: We use the Perpetual Inventory System in Ch6.

**Perpetual inventory system:**

- is used to demonstrate the accounting for inventory purchase and sale transactions.

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**Perpetual inventory system:**

- is used to demonstrate the accounting for inventory purchase and sale transactions.
- maintains the **inventory** and **CGS** on a perpetual basis.
- enables **inventory** and **CGS** to be updated every time inventory is bought or sold.
- is widely used in practice.

Note: The **periodic inventory system** requires the use of separate temporary accounts for PURCHASES, TRANSPORTATION, etc. This goes beyond the scope of this course.

# Journal entries for Inventory Purchase Transactions

1. Walmart receives $10,500 of bikes purchased on account.

2. When buying the bikes, Walmart must pay the transportation cost of $200 to a trucker (Cost Principle).

3. Walmart returned some of the bikes to the supplier and received a $500 reduction in the balance owed.

# Journal entries for Inventory Purchase Transactions

1. Walmart receives $10,500 of bikes purchased on account.

| | | |
|---|---|---|
| Inventory (+A) | 10,500 | |
| Accounts Payable (+L) | | 10,500 |

2. When buying the bikes, Walmart must pay the transportation cost of $200 to a trucker (Cost Principle).

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1. Walmart receives $10,500 of bikes purchased on account.

| | | |
|---|---|---|
| Inventory (+A) | 10,500 | |
| Accounts Payable (+L) | | 10,500 |

2. When buying the bikes, Walmart must pay the transportation cost of $200 to a trucker (Cost Principle).

| | | |
|---|---|---|
| Inventory (+A) | 200 | |
| Cash (-A) | | 200 |

3. Walmart returned some of the bikes to the supplier and received a $500 reduction in the balance owed.

# Journal entries for Inventory Purchase Transactions

1. Walmart receives $10,500 of bikes purchased on account.

| | | |
|---|---|---|
| Inventory (+A) | 10,500 | |
| Accounts Payable (+L) | | 10,500 |

2. When buying the bikes, Walmart must pay the transportation cost of $200 to a trucker (Cost Principle).

| | | |
|---|---|---|
| Inventory (+A) | 200 | |
| Cash (-A) | | 200 |

3. Walmart returned some of the bikes to the supplier and received a $500 reduction in the balance owed.

| | | |
|---|---|---|
| Accounts Payable (-L) | 500 | |
| Inventory (-A) | | 500 |

# Journal entries for Inventory Selling Transactions

4. Walmart sells four mountain bikes at a selling price of $200 per bike, for a total of $800 cash. The bikes had previously been recorded in Walmart's Inventory at a cost of $175 per bike, for a total cost of $700.

| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
|---|---|
| Cash | |
| Accounts Payable | |
| Inventory | |
| Revenues | |
| Cost of Goods Sold | |
| Gross Profit | |

# Journal entries for Inventory Selling Transactions

4. Walmart sells four mountain bikes at a selling price of $200 per bike, for a total of $800 cash. The bikes had previously been recorded in Walmart's Inventory at a cost of $175 per bike, for a total cost of $700.

| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

| | | |
|---|---|---|
| Cost of Goods Sold (+E, -SE) | 700 | |
| Inventory (-A) | | 700 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
|---|---|
| Cash | |
| Accounts Payable | |
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| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

| | | |
|---|---|---|
| Cost of Goods Sold (+E, -SE) | 700 | |
| Inventory (-A) | | 700 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
|---|---|
| Cash | 600 |
| Accounts Payable | |
| Inventory | |
| Revenues | |
| Cost of Goods Sold | |
| Gross Profit | |

# Journal entries for Inventory Selling Transactions

4. Walmart sells four mountain bikes at a selling price of $200 per bike, for a total of $800 cash. The bikes had previously been recorded in Walmart's Inventory at a cost of $175 per bike, for a total cost of $700.

| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

| | | |
|---|---|---|
| Cost of Goods Sold (+E, -SE) | 700 | |
| Inventory (-A) | | 700 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
|---|---|
| Cash | 600 |
| Accounts Payable | 10,000 |
| Inventory | |
| Revenues | |
| Cost of Goods Sold | |
| Gross Profit | |

# Journal entries for Inventory Selling Transactions

4. Walmart sells four mountain bikes at a selling price of $200 per bike, for a total of $800 cash. The bikes had previously been recorded in Walmart's Inventory at a cost of $175 per bike, for a total cost of $700.

| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

| | | |
|---|---|---|
| Cost of Goods Sold (+E, -SE) | 700 | |
| Inventory (-A) | | 700 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
|---|---|
| Cash | 600 |
| Accounts Payable | 10,000 |
| Inventory | 9,500 |
| Revenues | |
| Cost of Goods Sold | |
| Gross Profit | |

# Journal entries for Inventory Selling Transactions

4. Walmart sells four mountain bikes at a selling price of $200 per bike, for a total of $800 cash. The bikes had previously been recorded in Walmart's Inventory at a cost of $175 per bike, for a total cost of $700.

| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

| | | |
|---|---|---|
| Cost of Goods Sold (+E, -SE) | 700 | |
| Inventory (-A) | | 700 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
|---|---|
| Cash | 600 |
| Accounts Payable | 10,000 |
| Inventory | 9,500 |
| Revenues | 800 |
| Cost of Goods Sold | |
| Gross Profit | |

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| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

| | | |
|---|---|---|
| Cost of Goods Sold (+E, -SE) | 700 | |
| Inventory (-A) | | 700 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
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| | | |
|---|---|---|
| Cash (+A) | 800 | |
| Revenues (+R, +SE) | | 800 |

| | | |
|---|---|---|
| Cost of Goods Sold (+E, -SE) | 700 | |
| Inventory (-A) | | 700 |

**After the above transactions, by how much have the following accounts changed?**

| Account | Balance |
|---|---|
| Cash | 600 |
| Accounts Payable | 10,000 |
| Inventory | 9,500 |
| Revenues | 800 |
| Cost of Goods Sold | 700 |
| Gross Profit | 100 |

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1. Introduction to Inventory

2. Inventory System and Journal Entries
   - Periodic Inventory System
   - Perpetual Inventory System
   - Inventory Purchase Transactions
   - Inventory Selling Transactions

3. Multistep Income Statements and Profit Ratios

# Single-Step vs. Multi-Step Income Statements

**ABC, Inc. — Income Statement — For the Years Ended Dec 31** *(amounts in millions)*

**Single-Step**

| Account | 2027 | 2026 |
|---|---|---|
| **Revenues:** | | |
| Net Sales | 12,000 | 11,000 |
| **Expenses:** | | |
| Cost of Goods Sold | 7,800 | 7,200 |
| SG&A Expenses[1] | 2,000 | 1,900 |
| Other Expenses (Net)[2] | 100 | 50 |
| Income Tax Expense | 630 | 555 |
| **Net Income** | 1,470 | 1,295 |

**Multi-Step**

| Account | 2027 | 2026 |
|---|---|---|
| Net Sales | 12,000 | 11,000 |
| Cost of Goods Sold | 7,800 | 7,200 |
| Gross Profit | 4,200 | 3,800 |
| SG&A Expenses | 2,000 | 1,900 |
| Income from Operations | 2,200 | 1,900 |
| Other Expenses (Net) | 100 | 50 |
| Income before Income Tax | 2,100 | 1,850 |
| Income Tax Expense | 630 | 555 |
| Net Income | 1,470 | 1,295 |

[1] SG&A: Selling, General, and Administrative Expenses (e.g., salaries and wages, utilities, rent).

[2] Other Expenses (Net): items outside core operations (e.g., interest expense or revenue); beyond the scope of this course.

# Net Sales

- Some customers (e.g., 3%) return the products they purchased.
- Firms record the estimated returns as **Sales Returns and Allowances**.[4]
- Firm's **Net Sales** is the total sales revenue minus the sales returns and allowances.
- Income Statement usually starts with **Net Sales**, but it depends on the firm's policy.

**ABC, Inc.**
**Net Sales Calculation**
**For the Years Ended Dec 31**
*(amounts in millions)*

| **Account** | **2027** | **2026** |
|---|---|---|
| Sales Revenue | 12,300 | 11,200 |
| **Less: Sales Returns & Allowances** | 300 | 200 |
| **Net Sales** | **12,000** | **11,000** |
| Cost of Goods Sold | 7,800 | 7,200 |
| **Gross Profit** | 4,200 | 3,800 |
| ... | ... | ... |
| ... | ... | ... |
| **Net Income** | 1,470 | 1,295 |

[4] **Sales Returns and Allowances** is a contra account that is used to record the estimated returns of sales. This goes beyond the scope of this course. We do not cover Sales Returns and Allowances in detail and the related journal entries.

# Gross Profit Ratio

## Gross Profit

- A profitability measure that shows how much money is left over from revenues after accounting for the cost of goods sold.
- Calculated as: Net Sales – Cost of Goods Sold

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- A profitability ratio that measures how much **gross profit** is generated as a percentage of net sales.
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- A profitability ratio that measures how much **gross profit** is generated as a percentage of net sales.
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- **2027:** $\frac{4{,}200}{12K} = 0.35$ or 35%
- **2026:** $\frac{3{,}800}{11K} \approx 0.345$ or 34.5%
- **Interpretation:** For every \$1 of sales, about _____ cents is gross profit in 2027.

# Gross Profit Ratio

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# Net Profit Ratio

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- A profitability ratio that measures how much **net income** is generated as a percentage of net sales.
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- **2027:** $\frac{1{,}470}{12K} = 0.1225$ or 12.25%
- **2026:** $\frac{1{,}295}{11K} \approx 0.1177$ or 11.77%
- **Interpretation:** For every \$1 of sales, about _____ cents is net profit in 2027.

# Net Profit Ratio

## Net Profit

- A profitability measure that shows how much money is left over from revenues after accounting for all expenses.
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- **Interpretation:** For every \$1 of sales, about **12.25** cents is net profit in 2027.

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4 Appendix: FOB Shipping Terms (Not for exam)

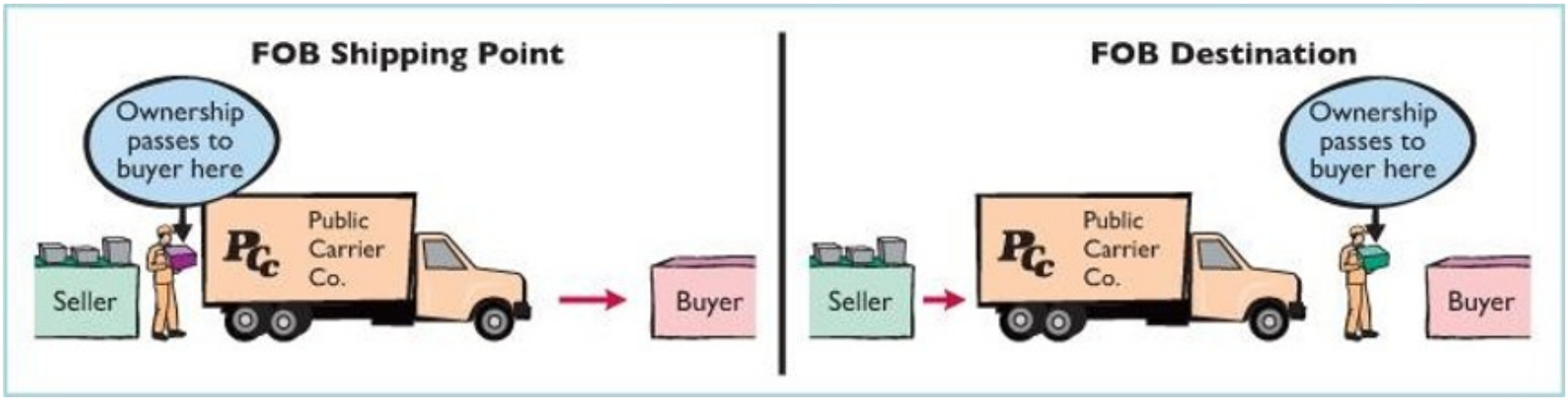
# FOB shipping terms

**FOB shipping point**

- Ownership transfers to the buyer when goods leave the **seller's location**.
- The **buyer** is responsible for shipping costs.

**FOB destination**

- Ownership transfers to the buyer when goods arrive at the **buyer's location**.
- The **seller** is responsible for shipping costs.



# Example - FOB Shipping Terms

**EX 1.** Walmart purchased $10,500 of bikes with **FOB destination**. Walmart doesn't pay the transportation cost.

| | | |
|---|---|---|
| Inventory (+A) | 10,500 | |
| Accounts Payable (+L) | | 10,500 |

**EX 2.** Walmart purchased $10,500 of bikes with **FOB shipping point**. Walmart must pay the transportation cost of $200 to a trucker (cash).

| | | |
|---|---|---|
| Inventory (+A) | 10,500 | |
| Accounts Payable (+L) | | 10,500 |
| Inventory (+A) | 200 | |
| Cash (-A) | | 200 |